Centralized Rineya Titan 0.6B: current research state

28 September 2026. **Data and measurement preparation have advanced; native E0 is blocked by host-RAM admission. No pretrained model has been loaded or trained in this campaign, no acquisition or retention effect has been measured, and no recipe has been promoted. The full research goal remains active and incomplete.**

Exact primary student and source

Primary: Rineya-Titan-Experimental-1.0.0-devfreeze, measured at **596,049,920 parameters in 310 BF16 tensors**. Geometry: 28 layers, hidden size 1024, MLP 3072, 16 query heads, 8 KV heads, head dimension 128, vocabulary 151936. Tied embeddings are declared; actual loaded tying remains an E0 gate. The declared 40960 context is not a measured operating envelope.

Weight SHA256: `537201676814ade103c4efa63f5ebe758ea309971803c4ad633dbca5b6b531b7`. Config SHA256: `660db3b73d788119c04535e48cf9be5f55bc3100841a718637ae695b442f27dd`. Tokenizer SHA256: `aeb13307a71acd8fe81861d94ad54ab689df773318809eed3cbe794b4492dae4`. Template SHA256: `a55ee1b1660128b7098723e0abcd92caa0788061051c62d51cbe87d9cf1974d8`.

Local source: `C:/Users/clclc/BusinessProject/Chat-Bubble/Rineya_Published_Cores/Rineya-Titan-Experimental-1.0.0-devfreeze`. Dense original Titan was chosen for the user's Titan preference and a simple BF16 starting point. No superiority over Titan-Sparse or Titan-Transition was measured. Their composition/precision factors remain outside the first causal chain; 344M is excluded from primary GPU work.

The remote parent was fetched and verified at `03db218ceebc61a2bc04f2482cf7e094045e01c1`. The execution branch `research/post-training-mimo-centralized-v1` was pushed at that ancestor before new implementation. No parent/main merge or history rewrite occurred. Existing untracked drafts were preserved under `evidence/superseded_initial_draft`; the initial checkout was not falsely reported fully clean. The complete supplied 66-page report was read; its SHA256 is `a1c5b5c66e1cef0edd04631ab7c487c93731b1a207bb67ee87fddd40cf43c2d7`. Its 344M resource estimates were not reused as Titan measurements.

Verified preparation

Artifact	Measured result	Scope	Remaining boundary
Primary binding	Exact files/header, 596049920 parameters	Static identity	Native loader/tie/parity unqualified
Acquisition	4096 TRAIN +256 selection +256 DEV parents	Independent deterministic verification	No student learning measured
Replay candidate	768 deterministic +256 curated language rows	Fixed sources, tokenized inputs	Current-source enrollment and broad preservation unqualified
E1 input/schedule plan	5120 sequences; 680532 nonpadding; 63546 supervised tokens; 256 steps	Frozen seed 101 schedule	No native executor or launch grant
Long-context DEV	32 worlds, 256 rows; exact 2024/4072 prompts	Tokenizer/visible-evidence mechanics	No model fit or preservation result
Language DEV/CAL	64 DEV +16 CAL worlds; 96 authored controls; max 166 prompt tokens	Candidate inputs and unassigned packets	Reference labels and grading unqualified
Software controls	43 new controls passed in 40.998 s, plus 2 fingerprint controls	Prior 172+89 runs retained separately	No pretrained efficacy implication

The maximum candidate TRAIN sequence is 398 tokens. Every target fits its registered cap without truncation. The shared non-thinking chat template and completion masks are bound; prior assistant turns in language conversations are context, and only the final response plus EOS is supervised. Source and derived file hashes are retained. Raw generated arrays remain local and reproducible from committed code and registrations; no model binaries were pushed.

Acquisition construction verified all 4608 open-row reference answers and rejected 4608 wrong-answer mutations in 17.009 s. Tokenizer profiling took 111.268 s. TRAIN contains 414554 nonpadding/49272 supervised tokens. CHECK has 512 acquisition parent hash reservations; FINAL remains unopened. Empty-set/text targets and repeated answers are disclosed: TRAIN has 105 empty sets and 12 empty strings; arithmetic has 428 distinct targets across 1024 parents. A TRAIN-mode constant-answer control scores DEV 2/64 arithmetic,1/64 text,3/64 sets,0/64 registers. These are data controls, not Titan scores.

Replay provenance and reference quality

AMENDMENT_002 froze 768 deterministic TRAIN rows over 640 semantic parents and 256 DEV rows over 192 parents. Answerability twins remain one parent. Retrieval, priority instructions and missing-evidence cases have separate solvers/verifiers. All 1024 open positives and wrong-answer controls passed. Construction took 14.495 s; tokenizer profiling 69.459 s. Deterministic TRAIN contributes 243336 nonpadding and 11814 supervised tokens; maximum open-role length 401, completion length 21.

Actual source inspection recovered all 14000 preservation rows by exact original TRAIN membership. It found 4000 v1 rows mislabeled v2 by the old generator. Original bytes were preserved; membership, not the defective label, determines origin. Quality exclusions left 4644 v2 candidates in 298 conservative template groups. Those groups are not asserted to be independent statistical parents. Frozen class quotas selected 256 language primaries and 768 bounded alternatives. Two assistant reviewers read all selected conversations: 250 references were acceptable and six had unsupported referents or claims. AMENDMENT_003 resolved five with reviewed same-group alternatives and one with the next fixed-order group in the same class. No target text was rewritten and no teacher target was authored. Original selections, flags and lineage are preserved. Curated language contributes 22642 nonpadding/2460 supervised tokens, maximum 122 total and 20 completion tokens. This is assistant source review, not human approval or calibrated native grading. The 256 language

TRAIN token arrays were subsequently materialized in 34.716 s, matching every previously measured token hash/count. No Torch/model call occurred.

A scoped audit compared 5632 open acquisition/deterministic-replay rows with 32800 historical whole-chat fingerprints and current normalized prompts. It found no collisions. It did not test historical user-only equality, all Titan ancestry, cross-schema semantic equivalence or pretraining exclusion. Protected payloads were not opened. No blanket decontamination claim follows.

Language candidate evaluation inputs

AMENDMENT_006 accepted data-only preparation of the proposed eight language strata. All 80 world specifications and their DEV/CAL roles were reviewed and committed at `5c2542b17613ed520de739dcc9e68b868a98143b` before conversations or replies were rendered. The resulting source package has 64 DEV worlds (32 single and 32 multi-turn), 16 separate CAL worlds, 64 illustrative DEV replies and 96 candidate CAL controls. The controls comprise two intended acceptable wordings and four primarily targeted defects per CAL world. Dimensions can interact; no qualified score vectors exist.

Two identified assistant reviewers read each other's complete source package. Before freeze, review added missing box/lid/car fit facts and changed one CAL decision structure that was too close to a DEV choice graph. Before export, one unsupported "naming vote" became "naming decision." Original drafts, hashes and reviews remain preserved. These are assistant source-quality checks, not human approval, qualified reference labels, 80 independent mechanisms or statistical scorer calibration.

Production at `c6d64603` exported the bank in 21.378 s and two unassigned JSON packet templates, each with 96 unique CAL controls and 16 hidden repeats. Rating views contain only opaque IDs, exact conversations/responses and instructions; coordinator mappings remain separate. Procedural separation does not guarantee blinding. Rating imports require freshly bound source rows, reject changed packets, retain missing/unjudgeable items and always remain diagnostic. No actual rating records or student responses exist.

Pinned tokenizer-only measurement took 34.202 s: 80 prompts total 8680 tokens, max 166; 160 exemplar/control completions total 5603 tokens, max 63 including EOS. All fit the 896/128 caps without truncation; 41 prior assistant turns are authored context. An artifact audit passed 214 checks with no findings, using separately written checks by the exporter author; it was not independent human review. The 43 new CPU controls passed in 40.998 s, plus two focused historical-fingerprint controls in 0.002 s.

The comparison against 42,000 original V1/V2 TRAIN rows found no full-context, system-stripped, normalized-context or normalized-user-sequence matches. Review of all 240 retrieved nearest contexts found no clear direct reuse within those pairs. A separate 32,800-entry historical whole-chat fingerprint comparison also found no matches and opened no protected payloads. These bounded checks do not establish complete Titan ancestry, semantic independence or unknown-pretraining exclusion. Comparison walls were 63.491 s and 1.215 s. Assistant authorship/review cost is not metered by these script receipts; it is not asserted to be free.

Exact long-context sentinel

AMENDMENT_004 prospectively accepted tokenizer-only construction after an independent blocked audit identified this remaining FAST task. Production at code commit `75e71793922a03e9dbc24588bf673fd8cbf202ab` completed in 139.455 s without Torch, model or GPU calls. DEV has 32 sampled numeric worlds, four per two-family/four- placement cell, and eight related rows per world. CHECK has 64 hash-only parent reservations, representing 512 future rows; no CHECK payload was rendered. FINAL 0 remains closed, not a selected confirmation population.

Each world crosses prompt length 2024/4072 with answerability and neutral/adversarial distractors. Thus 128 rows occupy each exact full-template length, totaling 780288 prompt tokens. A 24-token reserve gives total sequence envelopes 2048/4096. Targets are short symbols or UNKNOWN, measured at 1-4 tokens. No truncation or PAD/EOS filler is used. Required evidence spans are verified in body-relative native-token bands: early 10-20%, middle 45-55%, late 80-90%, or one block in each band when distributed. Paired rows preserve query, unchanged support text and controlled span positions.

The mechanisms are numeric interval routing and latest-authorized record selection. Each uses three necessary **record blocks**, not three atomic facts. The latest catalog has four entries, two per writer; authorization is independent of timeline order. Record/value order varies. Alias-only changes do not create new parents. The 32 worlds share only two structural mechanisms and do not establish 32 independent reasoning mechanisms. Each adversarial row includes different-key, obsolete ARCHIVE and unauthorized UNSIGNED shadows; their keys/payloads do not reveal the winner.

Pre-production review removed fixed-table-position, writer-count/time-order and decoy-key shortcuts. Independent reference and visible-text solvers verify all block ablations and causal interventions. A separate artifact audit found zero discrepancies in 15044 consistency checks, including 640 ablations, 64 causal controls and 288 decoy controls. It verified stored IDs/offsets internally, not by re-running the tokenizer; the native mapping is the separately bound production measurement. This bank qualifies measurement inputs, not student long-context capability.

Native failures and resource decision

E0 attempt 1 at `577ec6d5000bad11eaa72823d354b143b4178a26` failed before model load: Windows psutil rejected a pathlib.WindowsPath. The string-conversion repair preserved all thresholds. Its worker wall was 185.313 s; supervisor wall 191.422 s; peak process-tree RSS 705085440 bytes. A separate tokenizer preparation attempt selected an unrelated Hermes/Transformers 5.14.1 environment and failed its no-Torch guard before row processing. Exact failure duration is unmeasured, bounded 150-457 s; the incorrect progress interpretation is explicitly corrected in its receipt.

E0 attempt 2 at `285529d9ad328de49a3e820ebc8fbdb0d18d693e` measured 6.911331 GiB host available, below the 7.4 GiB post-import admission floor. GPU free 6.890625 GiB passed the 4 GiB admission floor. Worker RSS was 0.668579 GiB after CUDA setup, not pure import overhead. Worker/supervisor walls were 67.797/73.891 s; peak process-tree RSS 722173952 bytes. No pretrained load, update, training token or generated model token occurred. Later raw OS checks remained below 7.4 GiB even before imports, including 6.711 GiB on the third consecutive goal turn. No blind native retry was launched.

The pinned runtime is Python 3.11.15, Torch 2.10.0+cu126, Transformers 4.57.1, safetensors 0.7.0 and tokenizers 0.22.2 on an RTX 4070 Laptop with 8188 MiB. It has no removable sklearn/scipy/pandas/audio/vision dependency bundle; a minimal clone is not a substantiated memory fix. Existing guards remain 7.4 GiB admission and 6 GiB worker host memory, with external process-tree containment. This E0 path is now explicitly Windows-only pending qualification of another host's containment.

This rules out launching at the observed resource state. It does not prove that Titan cannot fit after admission, cannot learn or regresses. Model load/forward/ backward peaks, training/generation throughput and checkpoint/resume remain UNMEASURED. Historical FP32 DenseReference Titan scores are RECORDED FROM PRIOR WORK; they cannot be imported as current BF16 capability or source-correct membership.

Next decisive work and evidence limits

The next native experiment is unchanged E0 after a real resource-state change. The user was asked to free host memory. No paid cloud/teacher work, 344M fallback, guard reduction, CHECK/FINAL opening or automatic promotion has been authorized by a mechanics pass. Candidate recipe remains unchanged Titan with null acquisition and retention effects.

AMENDMENT_005 and production code at `676dcbabcd59247e97617e6eb7ad97fd5aa5af40` now bind all actual TRAIN arrays and a seed 101 plan: exactly 256 steps of 16 acquisition +4 replay rows, every row once, 13 warmup updates, final LR 0, checkpoints 0/64/128/192/256 and final-only selection. Prefix cursors bind schedule/data/config identity but do not establish restored native parameters, optimizer or RNG. The first plan used Python 3.13.2 and is preserved; the selected second plan uses pinned Python 3.11.15, which the future native worker needs for exact runtime-bound plan validation. Planning took 17.406/17.272 s respectively, with zero model calls or updates. A separate standard-library audit passed 68 checks with no findings, including actual TRAIN bytes, exact coverage, step math, cursor and source/config hashes. The two plans have identical row order and steps; only runtime metadata and its derived hashes differ. This audit qualifies the plan artifact, not a model run.

New preflight validates the E0 evidence bundle and always denies native launch. The response importer preserves failed attempts and parent-linked twins/eight-row worlds, but all imports remain structural or diagnostic. Callback claims cannot certify native capture, and language grading remains blocked. No actual responses have been enrolled. The 89 new CPU

controls test bindings, schedules, resume accounting, gate rejection and import mechanics; they are not a rerun of all 172 prior tests. There is still no runnable native E1 trainer or qualified current-source collector.

The next independent task is a concrete DEV input/scorer adapter. Short acquisition and retention token files contain reference completions after the prompt; a collector must bind and slice only the prompt positions. Long-context and language arrays have different prompt-only layouts. Bind actual source/token rows, parent groups and pure deterministic verifiers; keep targets separate from policy inputs. Do not invent a qualified runtime identity or numerical run budget to complete a generic manifest.

Language grading still needs assigned qualified reviewers, independent reference labels and calibration; source review cannot fabricate that authority. The 2048/4096 forward-only runtime extension, actual source enrollment and native E1 executor remain unqualified or absent. The 2 pp/5 pp preservation screens are exploratory, not noninferiority certificates. No preparation result earns native launch.

All judgments remain MEASURED / REPRODUCED, VERIFIED SOURCE, RECORDED FROM PRIOR WORK, INFERRED, PROPOSED, UNMEASURED or BLOCKED according to their actual evidence. Native training is blocked; the overall goal stays active because meaningful independent implementation work remains. See RESEARCH_LEDGER, RESULTS_MATRIX, RESOURCE_LEDGER, CHECKPOINT_LINEAGE and NEXT_EXPERIMENT for the durable audit trail.